

Lacoste and Romberg Model G Gravimeter How-To

WARNING: Be very careful with this instrument, as it is extremely delicate.

ALWAYS make sure it is clamped before moving it. Otherwise you will damage the spring, which is VERY expensive to repair.

How to use Lacoste-Romberg Gravimeter and convert meter readings to gravity:

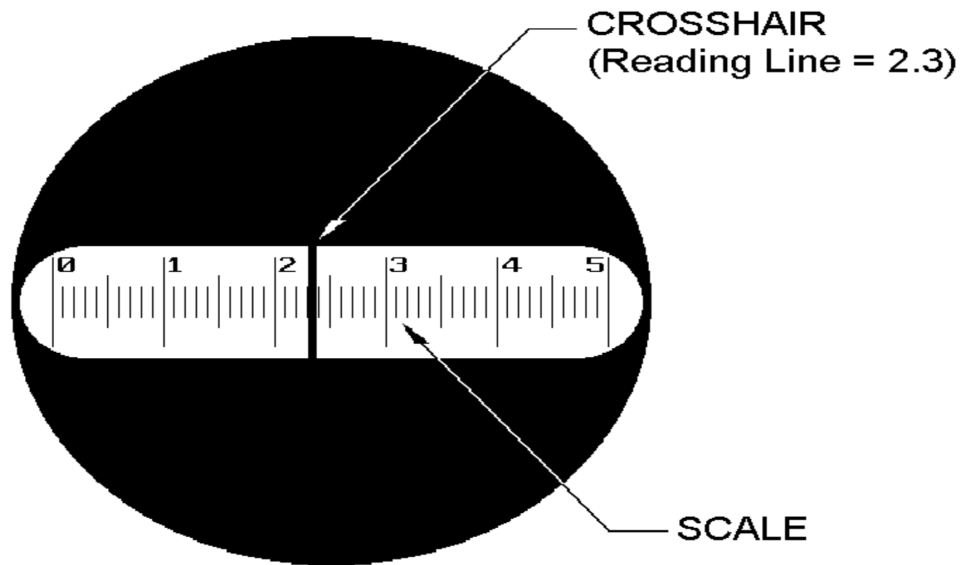
1. **Check that it is fully clamped.**
2. Level it.
3. Unclamp it.
4. Turn the dial until the left side of the cross hair just touches reading line 2.3 (see next page)
5. Read the number off the meter/dial as follows:
 - a. There is a 5 digit number, say 4269.5
 - b. The .5 can also be read from the thumb screw. The ticks mark an additional digit, say 4269.53
6. **Re-clamp the instrument.**
7. Convert the reading to mgal using the calibration chart (page 3)
 - a. Write the counter reading as 2 numbers, e.g: 4200 + 69.53
 - b. Find the first number in the table below under Counter Reading
 - c. From the table, get the "Value in milligals" $V=4432.96$
 - d. From the table, get the "Factor Interval" $FI=1.05750$
8. Final gravity is: $V + FI \cdot 69.53$, where 69.53 is the second number above. This is the number that is often saved for analysis.

To get the actual observed gravity in mgals, add 976000.

This will provide very precise measures of relative gravity, but the absolute value will be way off. To get absolute gravity, measure gravity at a site with known absolute gravity before and after your survey and correct your values to that value.

How to Read the Gravimeter, from Instruction Manual - Model G Meter

- The position of the beam is determined by the image of the crosshair in the microscope. The crosshair is a very fine wire attached to the beam. A reticle or scale is placed in the optical path for a reading reference. The total motion of the beam is 14 small scale divisions. The downscale or left side of the crosshair is used as the reading edge.



- Each meter has its characteristic reading line. There is a small placard on the meter lid indicating the reading line for the meter. In the example above, the reading line is 2.3.
- Bring the left side of the crosshair to the reading line by turning the nulling dial. If the crosshair needs to move to the right, turn clockwise. If it needs to move to the left, turn counterclockwise.
- Always approach the reading line from the same direction. For uniformity, we suggest approaching from left to right, (turning clockwise). If coming from the right side (counterclockwise), turn the dial about a quarter turn past the null and approach clockwise. The play or slack in the gears and universal joint would cause an error if the null is not approached always from the same direction.
- The number of turns of the nulling dial is limited by the counter. The counter is located under a window in the middle of the black lid and just to the right of the nulling dial. The maximum reading of the model G is 7000.

Calibration Chart from Instruction Manual for Lacoste and Romberg, Inc.

Model G Geodetic Gravity Meter No. 213

NOTE: Right hand wheel on counter indicates approximately one milligal

Counter Reading	Value in Milligals	Factor for Interval	Counter Reading	Value in Milligals	Factor for Interval
0	0	1.05439	3600	3798.52	1.05717
100	105.44	1.05427	3700	3904.24	1.05724
200	210.87	1.05419	3800	4009.96	1.05738
300	316.29	1.05413	3900	4115.69	1.05743
400	421.7	1.05419	4000	4221.44	1.05754
500	527.12	1.05406	4100	4327.19	1.05768
600	632.52	1.05405	4200	4432.96	1.0575
700	737.93	1.05404	4300	4536.74	1.05764
800	843.33	1.05405	4400	4644.53	1.05802
900	946.74	1.05406	4500	4760.33	1.05809
1000	1054.14	1.05409	4600	4850.14	1.05813
1100	1159.55	1.05414	4700	4961.9	1.05615
1200	1264.97	1.0542	4800	5067.77	1.05817
1300	1370.39	1.05427	4900	5713.59	1.05818
1400	1475.81	1.05434	5000	5279.6	1.05919
1500	1531.25	1.05443	5100	5385.22	1.05817
1600	1686.69	1.05454	5200	5490.04	1.05815
1700	1792.14	1.05468	5300	5596.85	1.05812
1800	1897.61	1.05485	5400	5702.67	1.05896
1900	2003.1	1.05503	5500	5808.47	1.05797
2000	2103.6	1.05522	5600	5914.37	1.0579
2100	2214.12	1.05532	5700	6020.06	1.0578
2200	2319.66	1.05551	5800	6125.84	1.05768
2300	2425.21	1.05563	5900	6231.61	1.06756
2400	2539.77	1.05574	6000	6337.36	1.06743
2500	2636.35	1.05583	6100	6443.11	1.05728
2600	2741.93	1.05593	6200	6548.83	1.05715
2700	2847.52	1.05606	6300	6654.55	1.067
2800	2953.13	1.05622	6400	6760.25	1.0568
2900	3058.75	1.0564	6500	6865.93	1.05656
3000	3164.39	1.05659	6600	6971.59	1.0563
3100	3270.05	1.05673	6700	7077.22	1.05602
3200	3375.72	1.05685	6800	7182.82	1.05572
3300	3481.41	1.05695	6900	7286.39	1.0554
3400	3587.1	1.05704	7000	7399.93	
3500	3692.81	1.05711			

Table for Recording Gravity Measurements (Group)

Floor #	Gravimeter reading